

Processing Pipelines

APPN Aerial SOP — Locked revision 1.02

Processing Pipelines

Important

This page documents the **standardised GRYFN processing pipelines** used within APPN for UAV-based data processing, intended for trained APPN staff processing CALViS and GOBI. Adherence to the pipelines below ensures reproducibility, quality assurance, and cross-project comparability. For any processing run that **deviates from the standard pipeline**, detailed records must be kept of every parameter changed, the rationale, and any anticipated implications for data quality.

This page documents the standardised GRYFN processing pipeline YAML files used within APPN for UAV-based data processing. These YAML configurations define consistent, transparent, and repeatable processing workflows across GRYFN-supported sensors and platforms, supporting reproducibility, quality assurance, and cross-project comparability. The files are intended to be used as reference and default templates for approved processing pipelines, with any deviations explicitly documented to maintain traceability and data integrity.

The standard pipelines and the data products they output are detailed below. For a tabular summary of output formats, resolutions and software compatibility, see Standard Data Products.

For additional information about all steps in this document, see the Gryfn Documentation

Document Structure

- CALViS — Standard Processing Pipeline
 - 1. Configure GPT defaults
 - 2. Format the raw data
 - 3. Create the grow bundle
 - 4. Load the Calvis_standard_pipeline
 - 5. Run the pipeline on a grow
- CALViS outputs
- GOBI — Standard Processing Pipeline
- GOBI outputs

CALViS — Standard Processing Pipeline

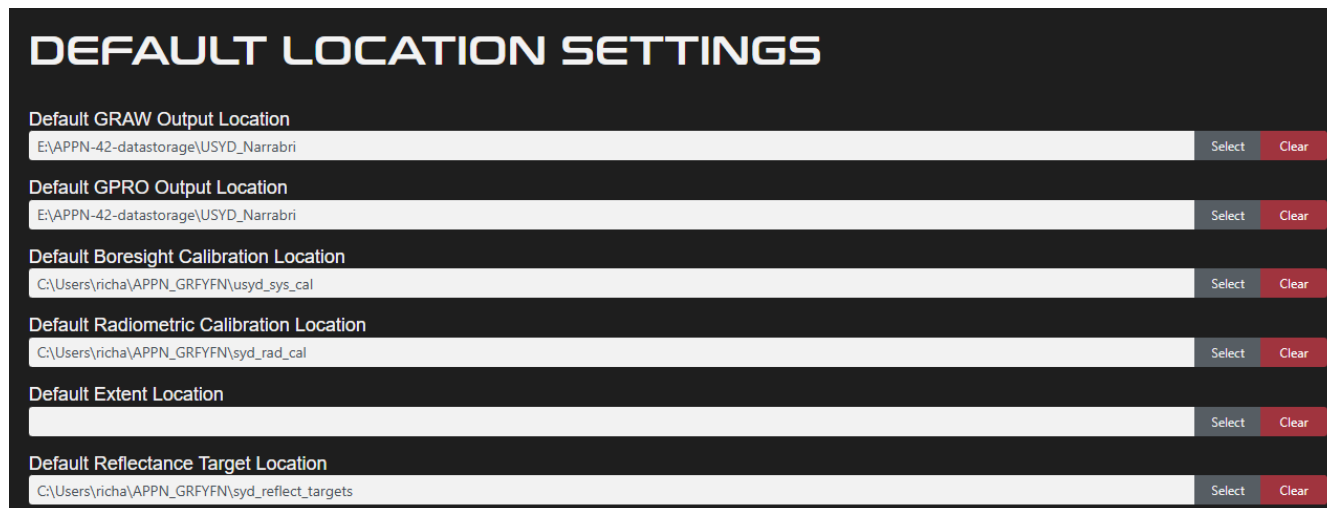
This section describes the standard CALViS processing workflow using the `Calvis_standard_pipeline_v1.0.yaml` file in the GRYFN Processing Tool (**GPT, v1.9.2**).

Note

The pipeline YAML lives in `Protocols/Pipelines/GryfnProcessingPipeline/yaml/`. Download the latest version from the link above before importing it into GPT (see Step 4).

1. Configure GPT defaults

Before processing any data, configure GPT so that it uses the correct radiometric calibration files and points to the correct panel target files.



Setting	Value	Select	Clear
Default GRAW Output Location	E:\APPN-42-datastorage\USYD_Narrabri		
Default GPRO Output Location	E:\APPN-42-datastorage\USYD_Narrabri		
Default Boresight Calibration Location	C:\Users\richa\APPN_GRFYFN\usyd_sys_cal		
Default Radiometric Calibration Location	C:\Users\richa\APPN_GRFYFN\syd_rad_cal		
Default Extent Location			
Default Reflectance Target Location	C:\Users\richa\APPN_GRFYFN\syd_reflect_targets		

Sensor identification (boresight calibration) Each CALViS unit has a unique sensor identifier embedded in the boresight calibration filename. The naming convention is:

Example: 20250527_cAHP-191_SystemCal.yml

Where 191 is the USYD sensor number; each CALViS unit is assigned its own unique number.

Important

When processing data, you **must** verify that the sensor number in the boresight calibration file matches the sensor used for data collection.

Radiometric calibration files Each sensor has an associated set of radiometric calibration files supplied by GRYFN. The radiometric calibration folder contains subfolders; it is recommended to remove the .nuc folder and keep the .raw folder to prevent the Gryfn Processing Tool from grabbing the wrong files.

Important

Verify that the **Radiometric Calibration Location** parameter in GPT points to the correct calibration files for your specific sensor(s).

Reflectance target values The Empirical Line Method (ELM) requires accurate reflectance target values for the calibration panels.

- Four (4) calibration panels are used for the ELM process.
- Each panel has specific, measured target reflectance values.
- These values are unique to your panel set.

Important

Ensure the **Reflectance Target Location** parameter references the correct target values that correspond to your specific calibration panels.

2. Format the raw data

The example CALViS flight used here is `run_00`, a flight from UOA.

A complete CALViS dataset has four key components — **VNIR**, **SWIR**, **LiDAR**, and **GNSS** — and these must be arranged correctly before a working graw can be bundled. The raw data folder will look like this:

2026_APEX > 2026Roseworthy-SA > CALVIS > 20260415 > run_00 > T0_raw >

Name	Date modified	Type	Size
100060_20260415_APEX1_AU_dark_2026_04_15_01_23_50	5/6/2026 3:42 PM	File folder	
100061_20260415_APEX1_AU_2026_04_15_01_24_28	5/6/2026 3:42 PM	File folder	
20260415_APEX1_AU_2026_04_15_01_24_11_297	5/6/2026 3:47 PM	File folder	
GNSS	5/6/2026 3:43 PM	File folder	
lidar	5/6/2026 3:47 PM	File folder	
Vault	5/6/2026 3:45 PM	File folder	

SWIR
VNIR
GNSS
Lidar

Note

When you download the VNIR data it has the **LiDAR** data nested inside the VNIR folder, and the **VNIR dark frames** are in the same folder. The **SWIR dark frames** are in a separate folder. The GNSS folder contains the relevant .T04 files.

3. Create the graw bundle

Pre-bundle checks Before each graw, confirm you have:

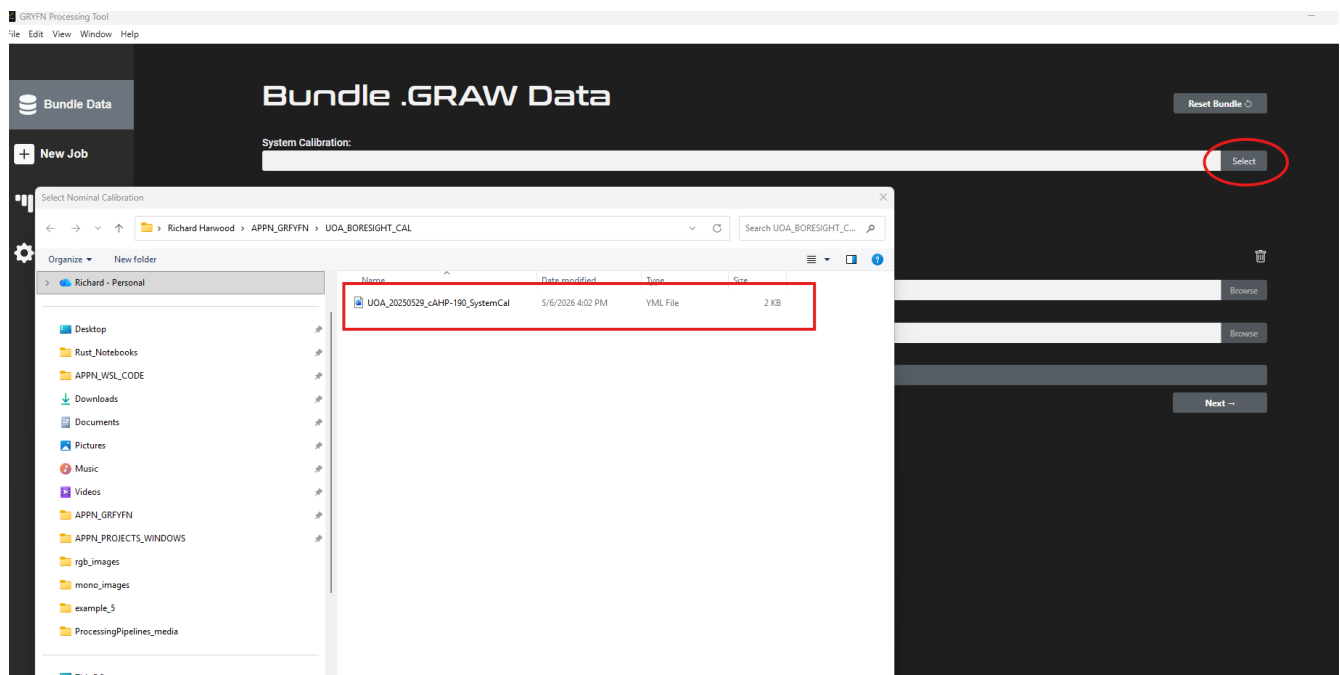
- ☐ System calibration file (matching the sensor used).
- ☐ Radiometric calibration files.
- ☐ Reflectance target file.

DEFAULT LOCATION SETTINGS

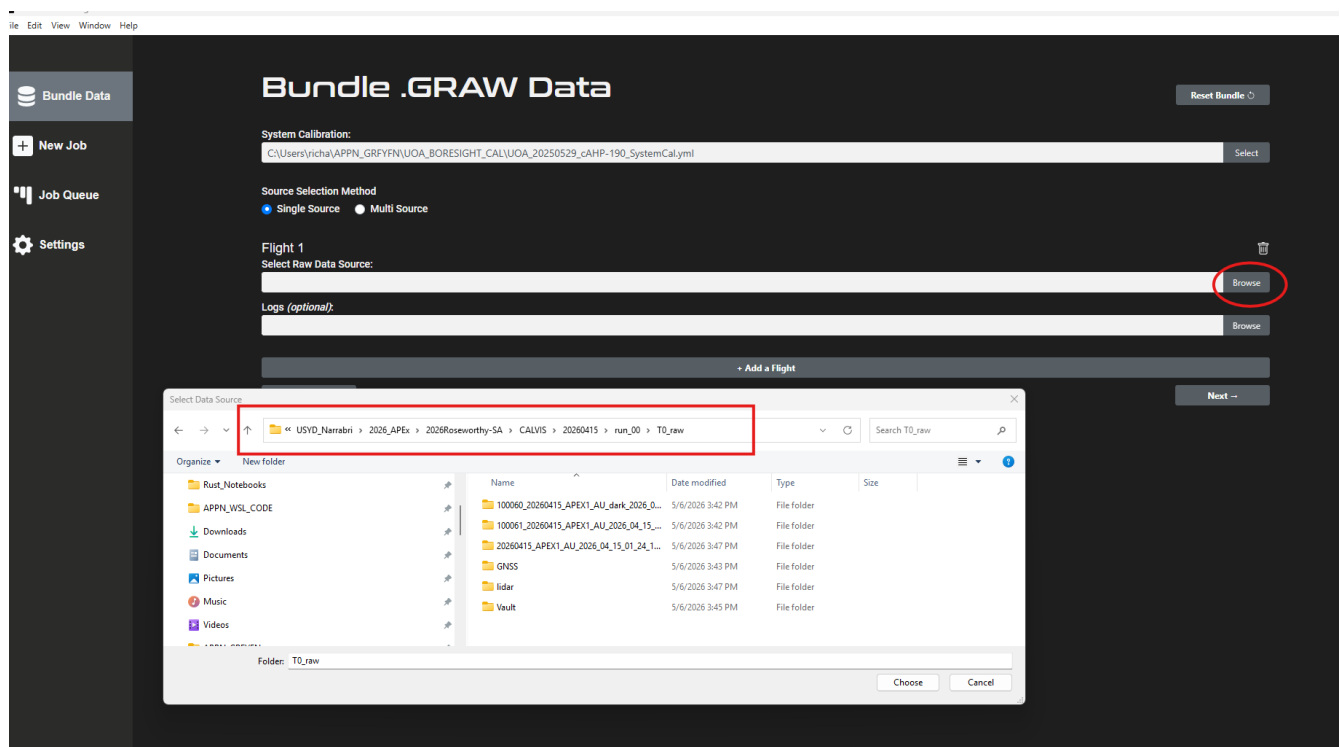
Default GRAW Output Location	E:\APPN-42-datastorage\USYD_Narrabri	Select	Clear
Default GPRO Output Location	E:\APPN-42-datastorage\USYD_Narrabri	Select	Clear
Default Boresight Calibration Location	C:\Users\richa\APPN_GRPYFN\UOA_BORESIGHT_CAL	Select	Clear
Default Radiometric Calibration Location	C:\Users\richa\APPN_GRPYFN\UOA_RAD_CAL	Select	Clear
Default Extent Location		Select	Clear
Default Reflectance Target Location	C:\Users\richa\APPN_GRPYFN\UOA_TARGETS_CAL	Select	Clear

Bundling steps

1. Set the system calibration file to match the sensor you are using.



2. Choose the raw data path.



3. Click *Next* to view the optional parameters.

File Edit View Window Help

Bundle .GRAW Data

Optional Parameters

Processing Extent:

Pilot Selection Method

Pilot:

Location:

Conditions:

Reset Bundle

Back Next

Note

Unless you have a specific reason to set an extent, **leave it blank** — data can be cropped to the hyperspectral capture area later, which is ideal for most standard flights. Other fields here are also optional.

- Click **Next**. Give the bundle a logical name and save it under T0_raw.

Bundle .GRAW Data

Reset Bundle

Name Your Bundle:

Select Folder to Store Bundle:

Back Create Bundle

- Click **Create bundle**. You should see a progress report while the bundle is being created.

JOB QUEUE

Bundling: UOA_APEX_run_00

UOA_APEX_run_00

Completing step: 36 of 126

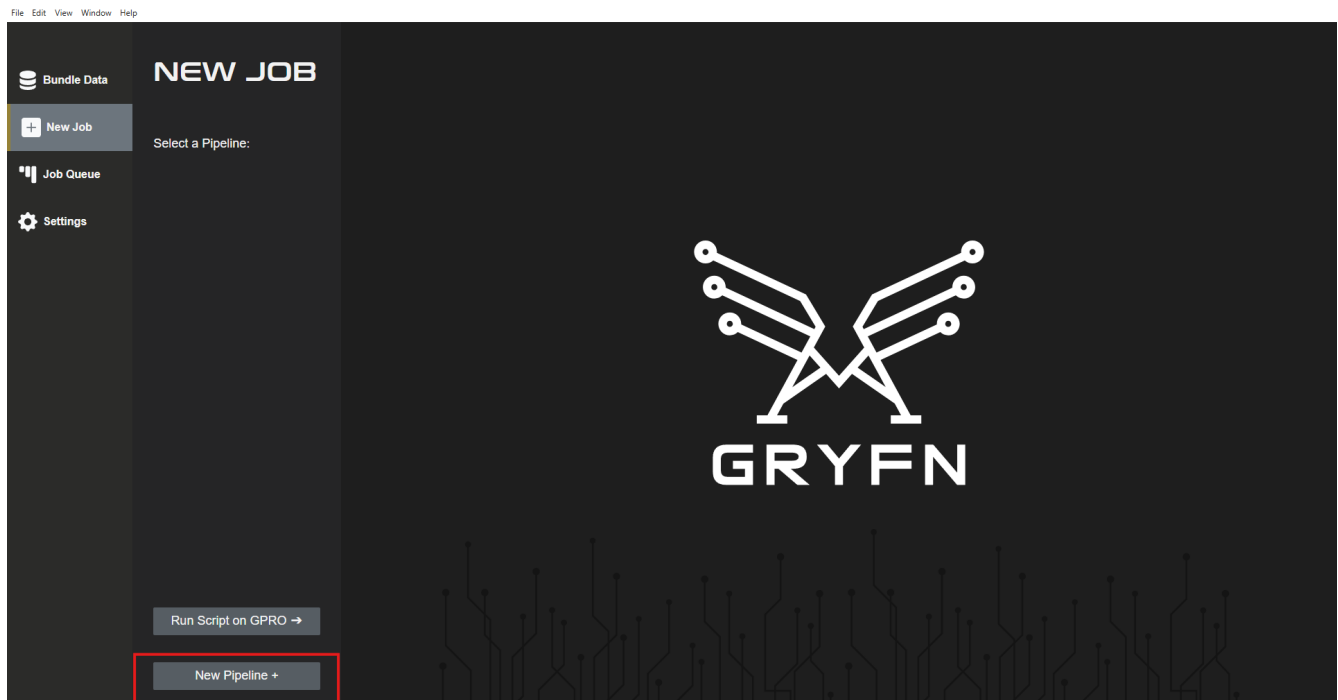
Time Elapsed: 00:00:39
Current Status: Bundling VNIR Data.
Processing Log:

```
[123/126] 2026/05/06 16:13:43 440929 INFO Copying 20260415_APEX1_AU_2026_04_15_01_24_11_297\FrameIndex_9870.txt.
[23/126] 2026/05/06 16:15:45 4459213 INFO Copying 20260415_APEX1_AU_2026_04_15_01_24_11_297\FrameIndex_9870.txt.
[25/126] 2026/05/06 16:15:45 470988 INFO Copying 20260415_APEX1_AU_2026_04_15_01_24_11_297\gpsMonitor.json.
[26/126] 2026/05/06 16:15:45 470987 INFO Copying 20260415_APEX1_AU_2026_04_15_01_24_11_297\imu_gps.txt.
[27/126] 2026/05/06 16:15:45 524671 INFO Copying 20260415_APEX1_AU_2026_04_15_01_24_11_297\raw_0.bin.
[27/126] 2026/05/06 16:15:48 678133 INFO Copying 20260415_APEX1_AU_2026_04_15_01_24_11_297\raw_0.hdr.
[28/126] 2026/05/06 16:15:48 682984 INFO Copying 20260415_APEX1_AU_2026_04_15_01_24_11_297\raw_1000.bin.
[30/126] 2026/05/06 16:15:55 268344 INFO Copying 20260415_APEX1_AU_2026_04_15_01_24_11_297\raw_1000.hdr.
[30/126] 2026/05/06 16:15:55 272511 INFO Copying 20260415_APEX1_AU_2026_04_15_01_24_11_297\raw_10874.bin.
[31/126] 2026/05/06 16:16:04 423549 INFO Copying 20260415_APEX1_AU_2026_04_15_01_24_11_297\raw_10874.hdr.
[33/126] 2026/05/06 16:16:04 427844 INFO Copying 20260415_APEX1_AU_2026_04_15_01_24_11_297\raw_11874.bin.
[34/126] 2026/05/06 16:16:15 790948 INFO Copying 20260415_APEX1_AU_2026_04_15_01_24_11_297\raw_11874.hdr.
[35/126] 2026/05/06 16:16:15 795515 INFO Copying 20260415_APEX1_AU_2026_04_15_01_24_11_297\raw_12874.bin.
[36/126] 2026/05/06 16:16:22 631682 INFO Copying 20260415_APEX1_AU_2026_04_15_01_24_11_297\raw_12874.hdr.
[37/126] 2026/05/06 16:16:22 636280 INFO Copying 20260415_APEX1_AU_2026_04_15_01_24_11_297\raw_13852.bin.
```

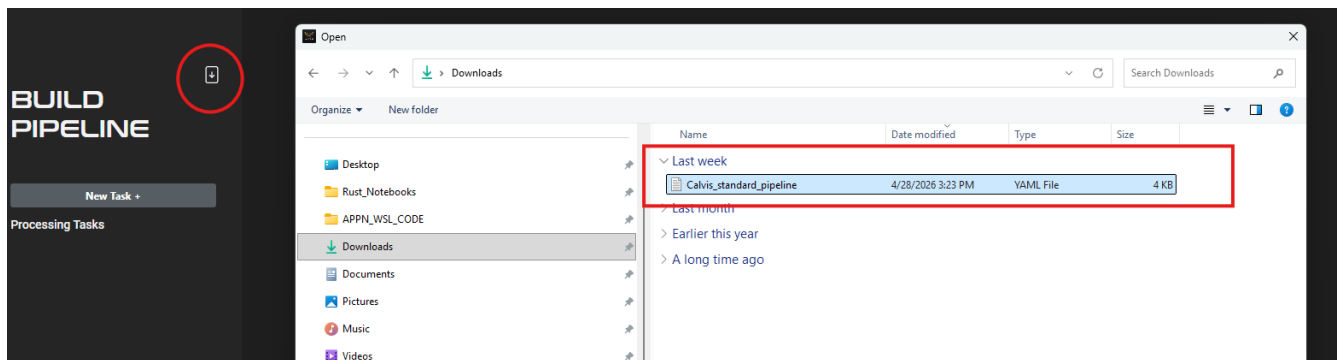
4. Load the Calvis_standard_pipeline

Import the standard CALViS pipeline YAML (Calvis_standard_pipeline_v1.0.yaml) into GPT before running a job for the first time:

1. Open the pipelines manager.



2. Import the Calvis_standard_pipeline YAML.



3. Confirm the imported pipeline is listed.



5. Run the pipeline on a graw

1. Select **New Job** and choose **Calvis_std** from the pipelines list. Choose your graw, name your gpro, and set the gpro output location.



Important

The gpro **must** be saved under T1_proc.

2. Configure GNSS processing. We currently recommend **PPRTX**.

General

GNSS

LIDAR

VNIR

SWIR

+ Add Script

GNSS: APX-15Lv3

Flight 1

Force Reprocess ☒

First POS File 6241C03151202604150124.T04 Select

Last POS File 6241C03151202604150133.T04 Select

GNSS Mode PP-RTX ▾

Datum GDA2020 ▾

Grid Australia (MGA2020) ▾

Zone MGA2020 MGA zone 54 ▾

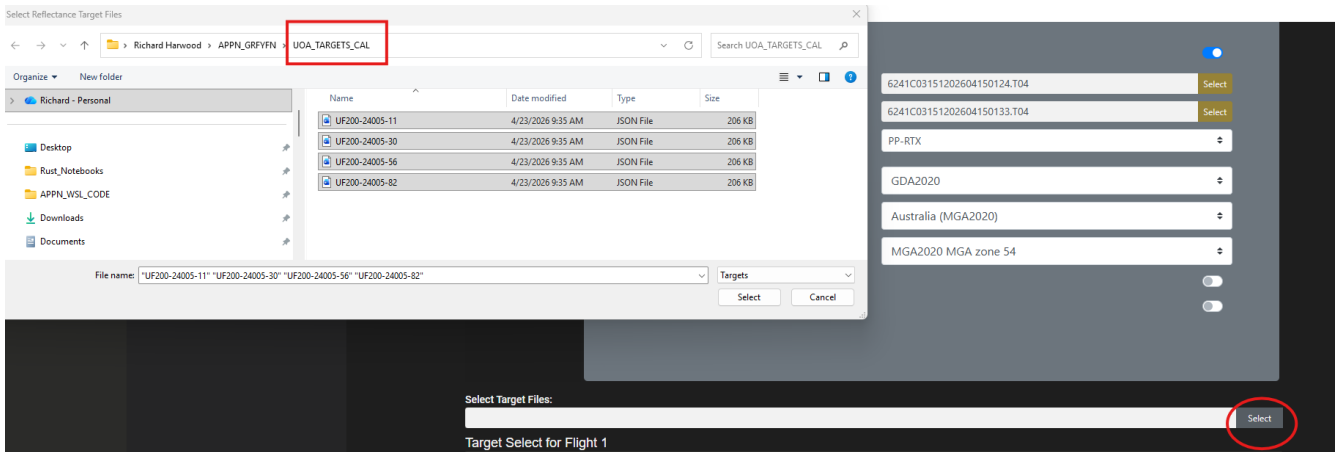
Realtime ☐

Override GNSS/IMU Lever Arms ☐

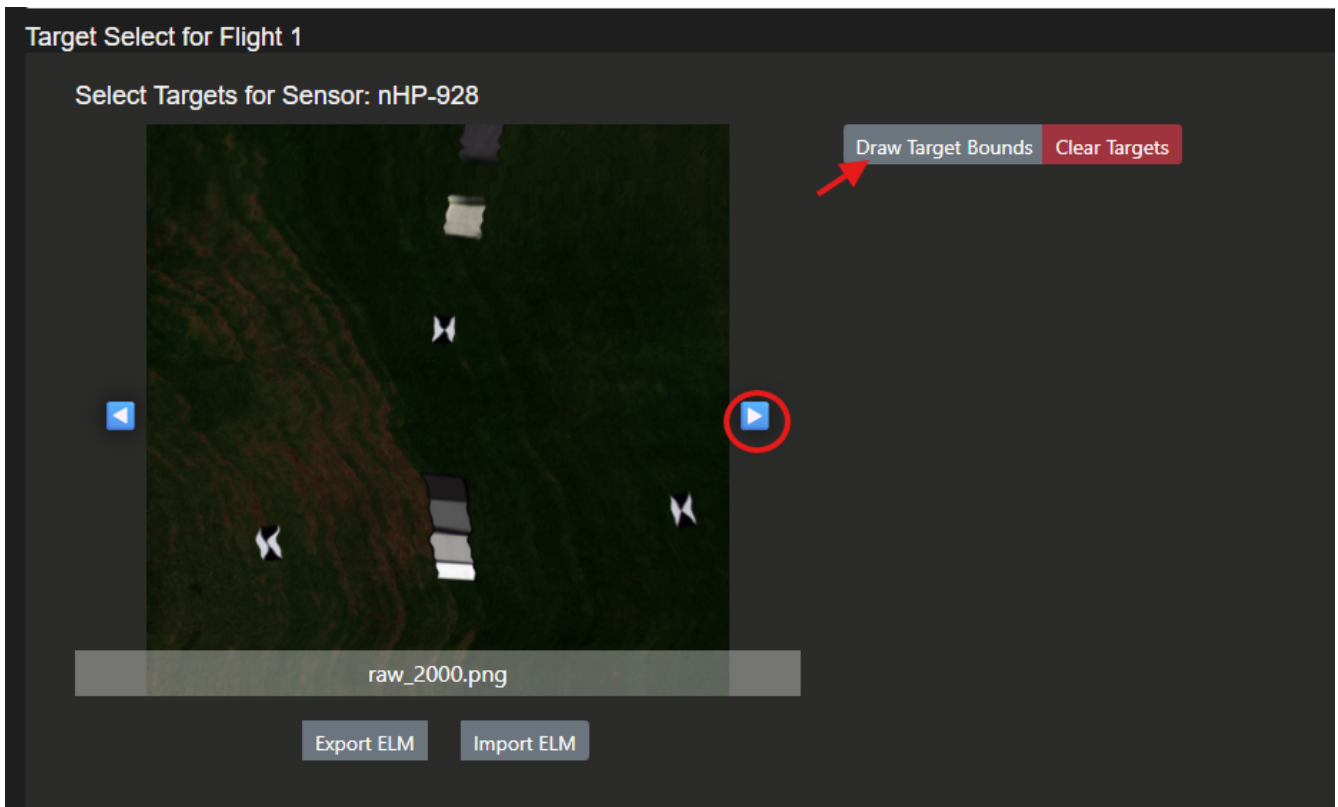
Caution

Double-check your **Datum**, **Grid**, and **Zone** before continuing. Errors here propagate through every downstream product.

- Load the reflectance target files. For this example flight we use the UOA panel target files.



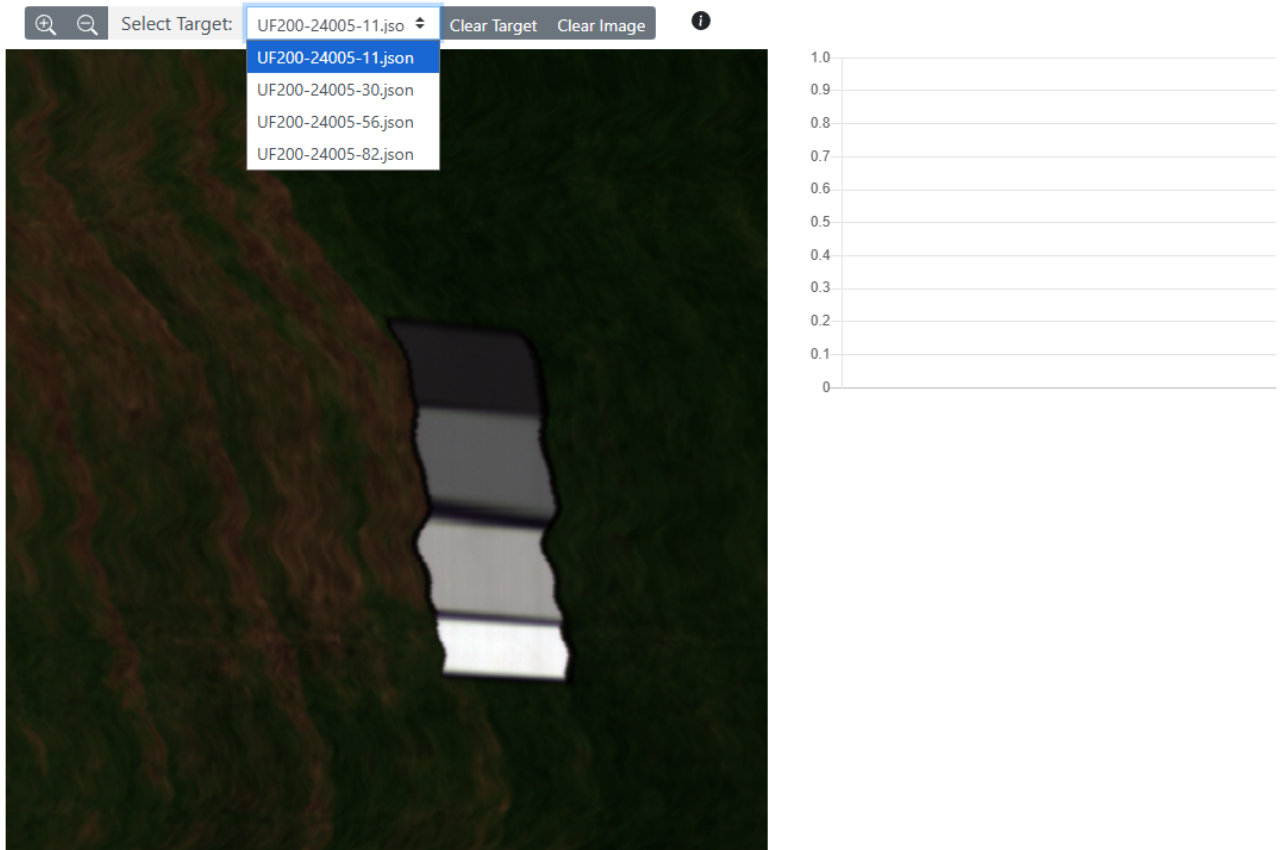
- VNIR ELM.** Cycle through the images until you find the four panels, then click **Draw target bounds**.



Choose a target to work on:

raw_2000.png ← →

Done



Hold right-click and drag a rectangle around the panel:

raw_2000.png

Done

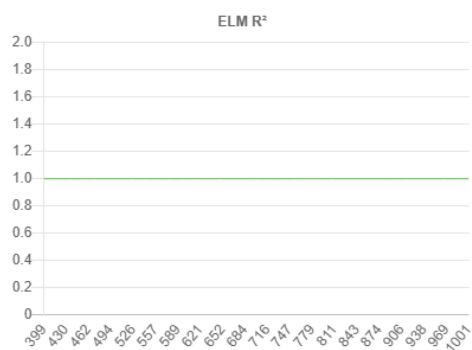
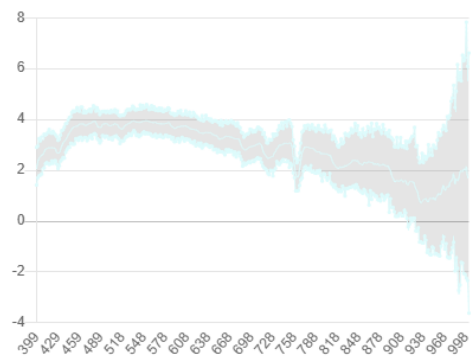


Select Target:

UF200-24005-11.iso

Clear Target

Clear Image

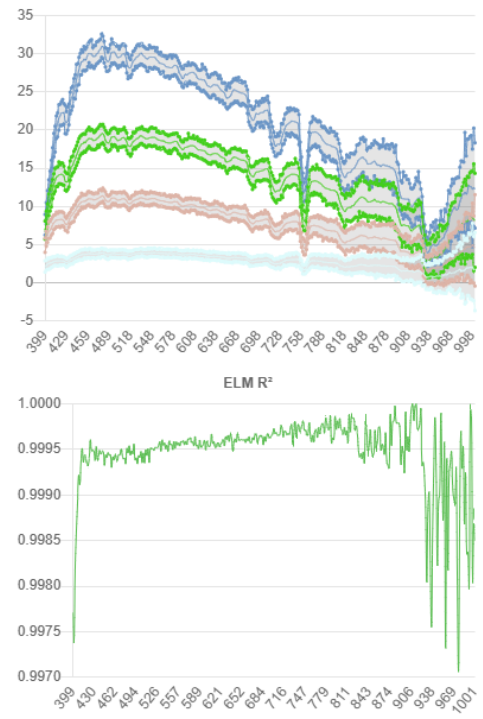


Repeat for the remaining three panels:

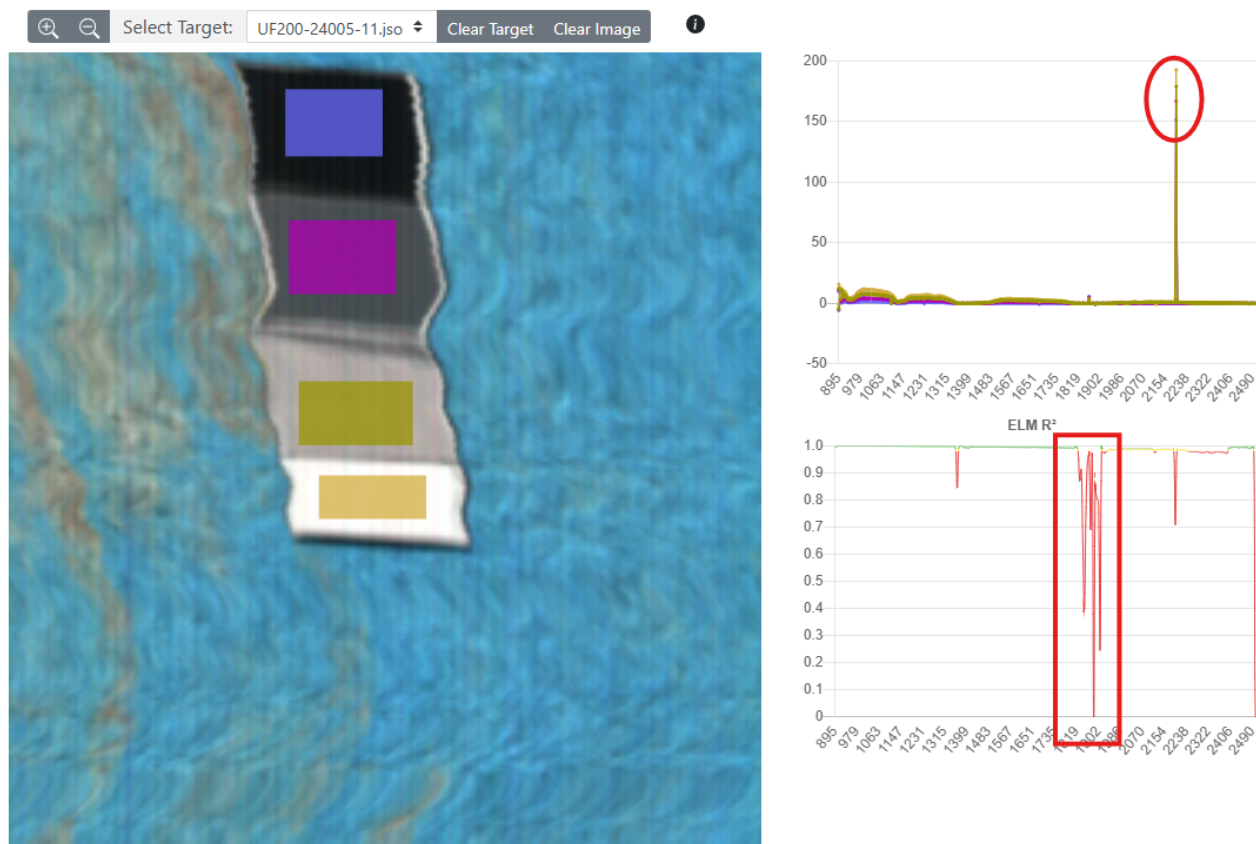
raw_2000.png

Done

Select Target: UF200-24005-82.jsa Clear Target Clear Image



5. **SWIR ELM.** The process is the same as VNIR.



Tip

The SWIR will have water bands (red box in the figure above) and sometimes rogue values (red circle). As a rule of thumb, **draw normal-sized boxes that cover the panel** rather than cherry-picking small areas to get a neater ELM.

[!IMPORTANT] make sure to note that the Y-axis in the top right box is showing radiance units and not raw counts (Should be obvious, ranges in the tens/hundreds vs thousands/tens of thousands)]

6. Click **Submit**.

CALViS outputs

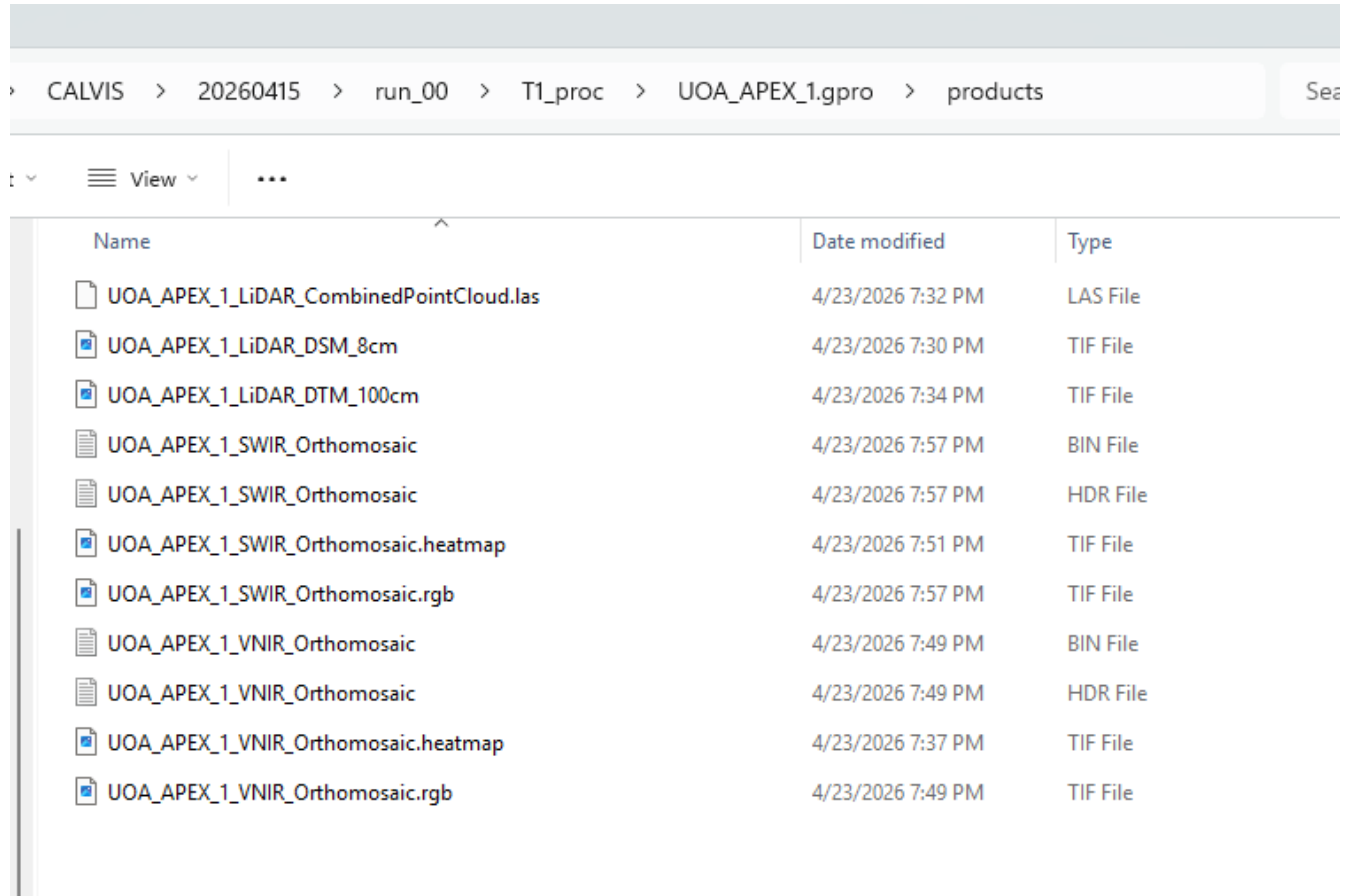
The CALViS standard pipeline produces the LiDAR and hyperspectral products listed below. See Standard Data Products for the canonical specifications, file sizes, and software compatibility.

Product	Output filename	Resolution	Format	Notes
LiDAR Digital Surface Model (DSM)	LiDAR_DSM.tif	8 cm (fixed)	GeoTIFF	Extent: VNIR scene
LiDAR Digital Terrain Model (DTM)	LiDAR_DTM.tif	1 m	GeoTIFF	Extent: processing extent
Combined LiDAR Point Cloud	LiDAR_CombinedPointCloud.las/.laz	Native point spacing	LAS	Outliers removed during combination
VNIR Orthomosaic	VNIR_Orthomosaic.bin	4 cm (GSD-based)	ENVI (.bin + .hdr)	binning = 2, radiometric calibration applied
SWIR Orthomosaic	SWIR_Orthomosaic.bin	4 cm (GSD-based)	ENVI (.bin + .hdr)	binning = 2, radiometric calibration applied

Inspecting the CALViS Outputs.

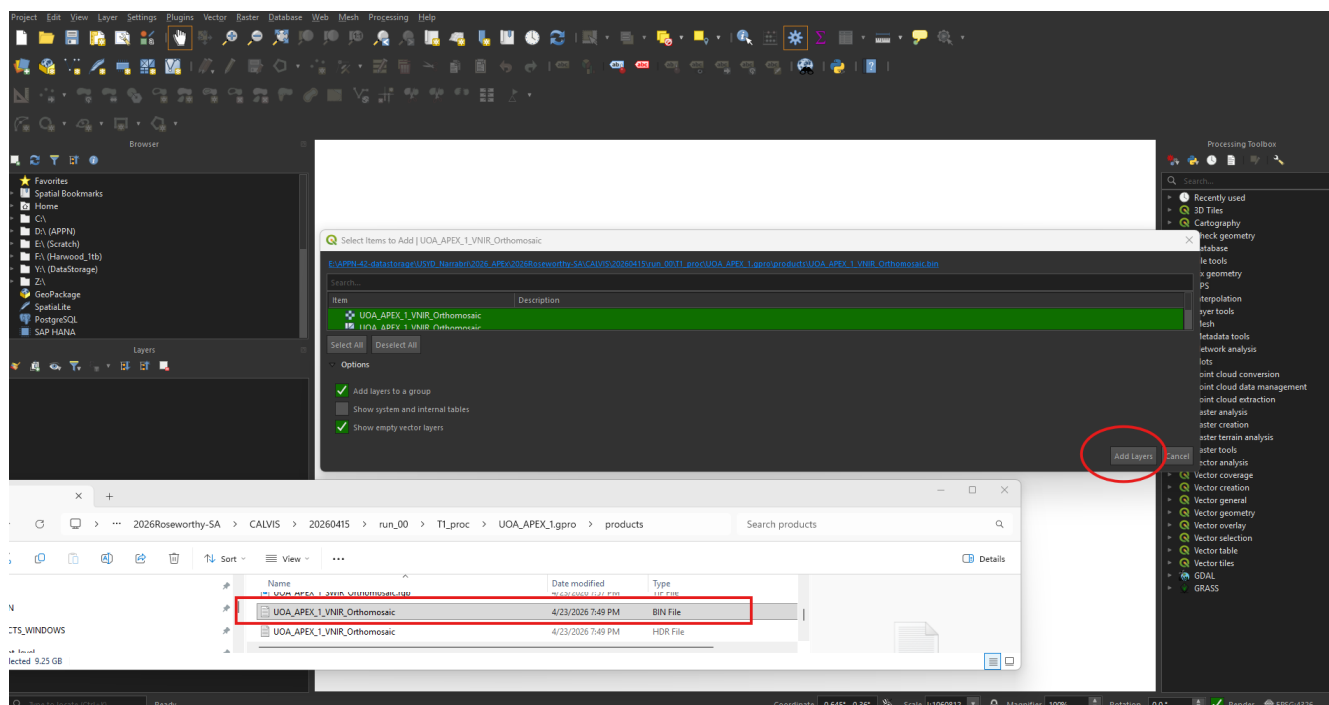
To inspect the DSM, DTM, RGB orthomosaic and the VNIR orthomosaic we recommend QGIS <https://qgis.org/>, shown below is QGIS v4.0 +, to view the pointcloud we recommend CloudCompare <https://www.cloudcompare.org/>. Note that the same approach works for the GOBI products, the only difference being the RGB orthomosaic which can also be viewed in QGIS.

The data outputs are stored in the "products" folder in the gpro

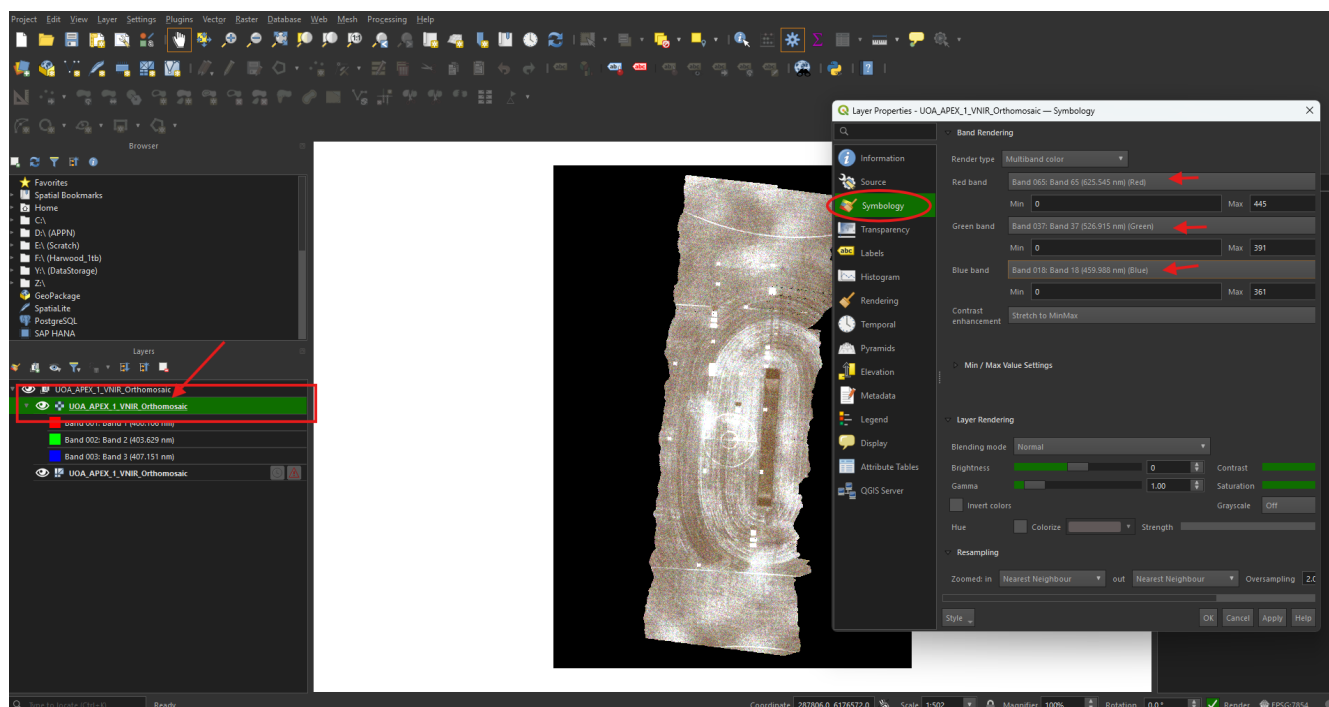


Name	Date modified	Type
UOA_APEX_1_LiDAR_CombinedPointCloud.las	4/23/2026 7:32 PM	LAS File
UOA_APEX_1_LiDAR_DSM_8cm	4/23/2026 7:30 PM	TIF File
UOA_APEX_1_LiDAR_DTM_100cm	4/23/2026 7:34 PM	TIF File
UOA_APEX_1_SWIR_Orthomosaic	4/23/2026 7:57 PM	BIN File
UOA_APEX_1_SWIR_Orthomosaic	4/23/2026 7:57 PM	HDR File
UOA_APEX_1_SWIR_Orthomosaic.heatmap	4/23/2026 7:51 PM	TIF File
UOA_APEX_1_SWIR_Orthomosaic.rgb	4/23/2026 7:57 PM	TIF File
UOA_APEX_1_VNIR_Orthomosaic	4/23/2026 7:49 PM	BIN File
UOA_APEX_1_VNIR_Orthomosaic	4/23/2026 7:49 PM	HDR File
UOA_APEX_1_VNIR_Orthomosaic.heatmap	4/23/2026 7:37 PM	TIF File
UOA_APEX_1_VNIR_Orthomosaic.rgb	4/23/2026 7:49 PM	TIF File

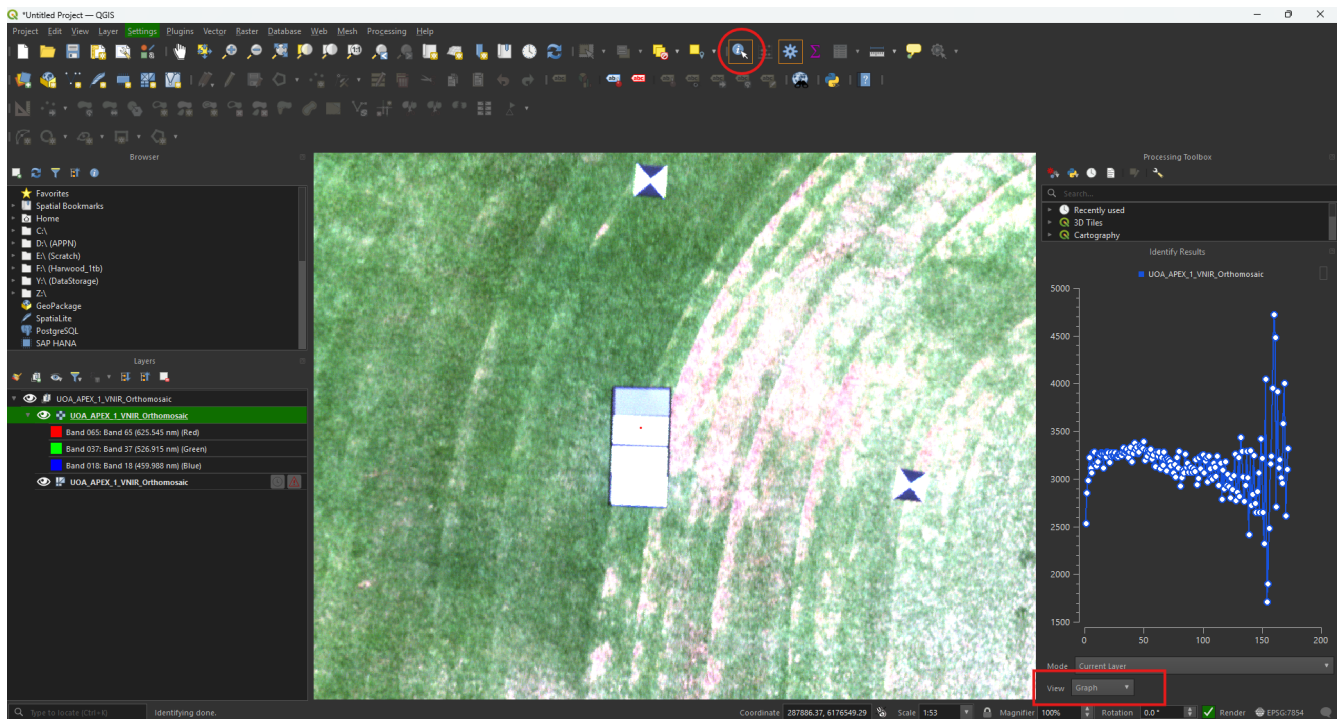
Inspecting the VNIR orthomosaic Open QGIS 4 and drag and drop the VNIR ".bin" then select "add layers"



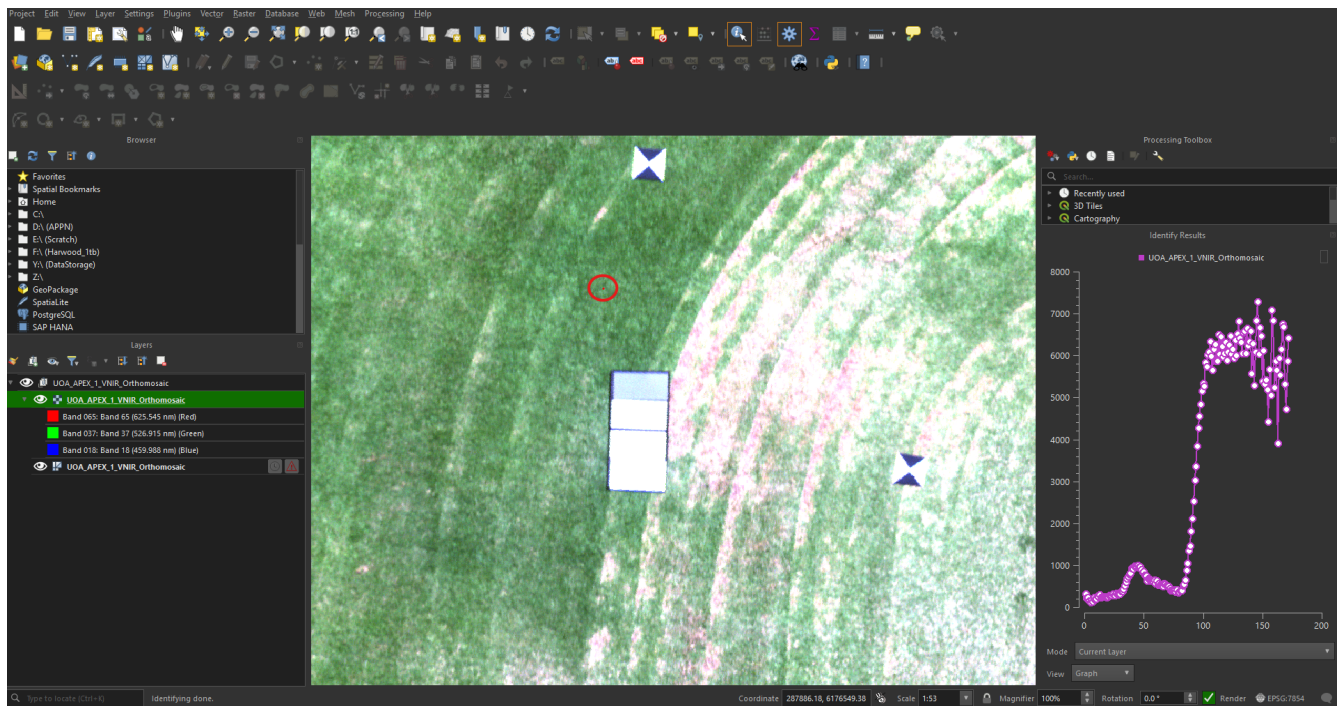
You will notice that the hyperspectral orthomosaic has a poor visualisation, this is because the default bands (wavelengths) visualised in QGIS are not ideal. We recommend setting the visualisation to Red Green Blue. To do this double click the file in the “Layers” section of QGIS, navigate to symbology and set the Red Green and Blue bands to the appropriate colour.



Next, it can be helpful to investigate, and sanity check some reflectance values. To do this we can use the “Identify Features” button. Click the button and then click a pixel on the orthomosaic. A good sanity check is to use a GRYFN panel, for example click on the 30% panel and check that the values are hovering around the 30% reflectance. One you click a pixel it defaults to a table, switching it to “graph” can make it easier to quickly interpret.

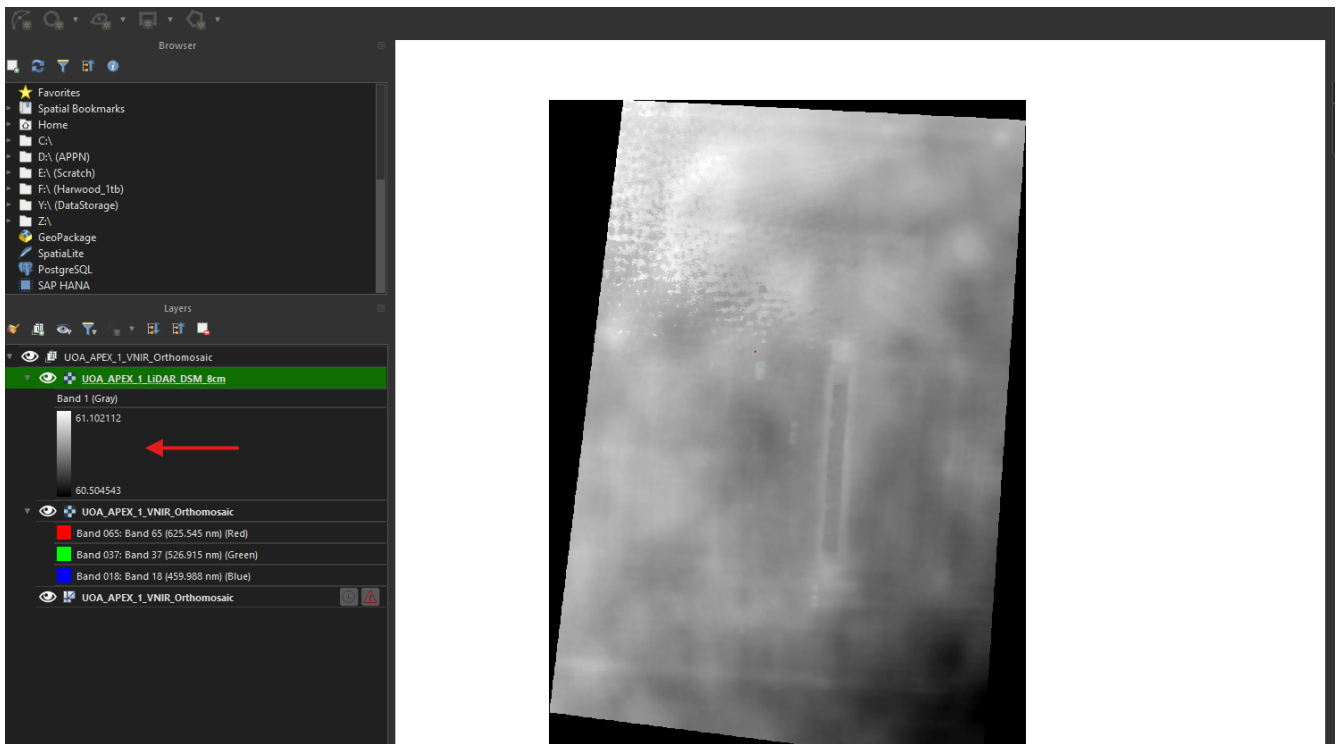


Following this you can click on grass, a crop, whatever is of interest to your flight (below is grass)

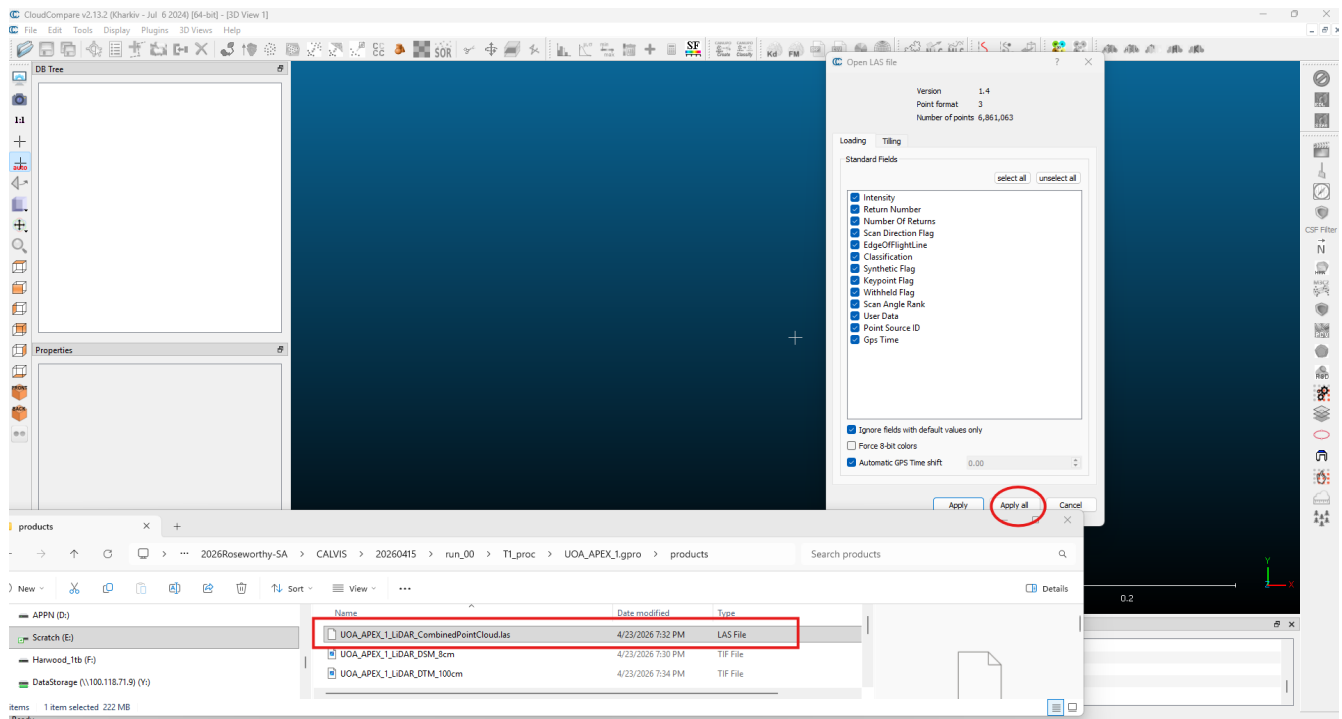


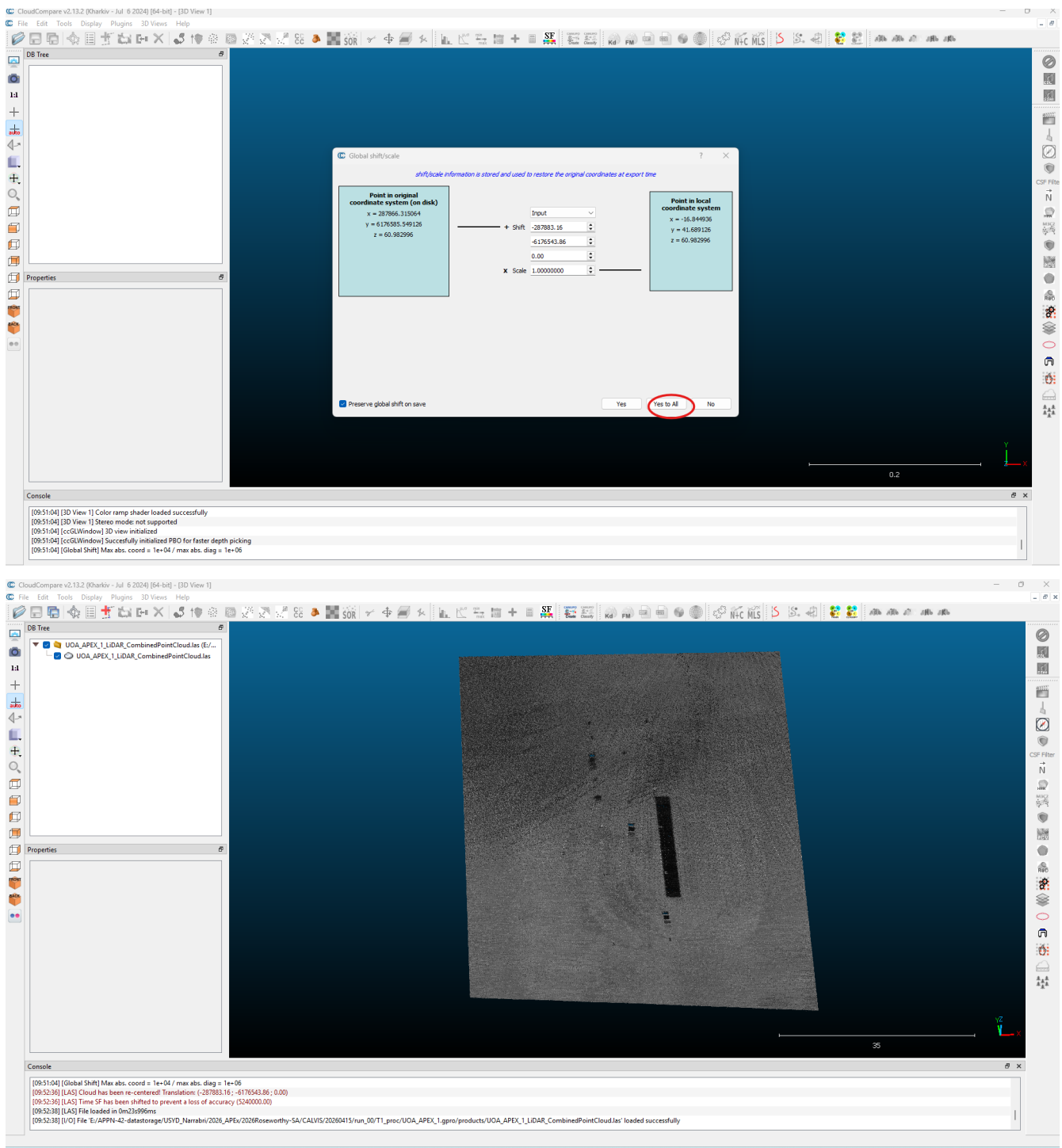
The steps for the SWIR are the same except there are no presets for RGB bands.

If your VNIR Orthomosaic looks odd, it can be good to check your DSM to make sure the LIDAR was working. To investigate your DSM just drag and drop the “.TIF” file into QGIS (note that this is actually a GeoTiff and contains spatial information). Make sure that the min max values make logical sense (e.g. the distance between the ground and the tallest feature)



Lastly, if you want to view your point cloud you can drag and drop the (.las or .laz file) into CloudCompare





All these steps apply to the GOBI the only difference being the RGB image (which is a Geotiff and loads just like the DSM)

GOBI — Standard Processing Pipeline

The GOBI standard processing pipeline is defined by `Gobi_standard_pipeline_v1.0.yaml`.

Note

The GOBI workflow closely mirrors the CALViS walkthrough above; only the GOBI-specific differences are called out below.

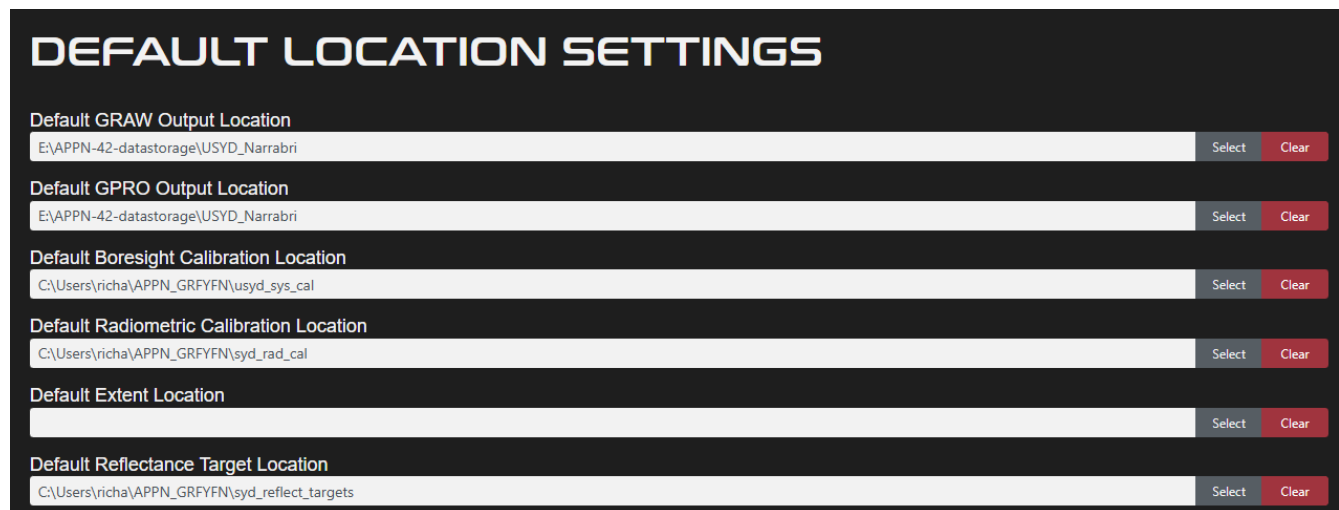
[!IMPORTANT] You need to set RGB resolution based on your flight height (you can get this from Gryfn Flight Calculator). The value in this pipeline is a placeholder only. In future we will work towards a pipeline for a suite of popular altitudes with an appropriate RGB resolution value.

The GOBI processing workflow is very similar to the CALViS. The main difference is that there is no SWIR but there is RGB data.

1. Configure GPT defaults

Repeated to make sure double checking is done

Before processing any data, configure GPT so that it uses the correct radiometric calibration files and points to the correct panel target files.



DEFAULT LOCATION SETTINGS

Default GRAW Output Location	E:\APPN-42-datastorage\USYD_Narrabri	Select	Clear
Default GPRO Output Location	E:\APPN-42-datastorage\USYD_Narrabri	Select	Clear
Default Boresight Calibration Location	C:\Users\richa\APPN_GRFYFN\usyd_sys_cal	Select	Clear
Default Radiometric Calibration Location	C:\Users\richa\APPN_GRFYFN\syd_rad_cal	Select	Clear
Default Extent Location		Select	Clear
Default Reflectance Target Location	C:\Users\richa\APPN_GRFYFN\syd_reflect_targets	Select	Clear

Sensor identification (boresight calibration) Each GOBI unit has a unique sensor identifier embedded in the boresight calibration filename. The naming convention is:

Example: 20241008_GOBI2405-APPN3_SystemCal.yml

Where APPN3 is the USYD sensor number; each GOBI unit is assigned its own unique number.

Important

When processing data, you **must** verify that the sensor number in the boresight calibration file matches the sensor used for data collection.

Radiometric calibration files Each sensor has an associated set of radiometric calibration files supplied by GRFYN.

Important

Verify that the **Radiometric Calibration Location** parameter in GPT points to the correct calibration files for your specific sensor(s).

Reflectance target values The Empirical Line Method (ELM) requires accurate reflectance target values for the calibration panels.

- Four (4) calibration panels are used for the ELM process.

- Each panel has specific, measured target reflectance values.
- These values are unique to your panel set.

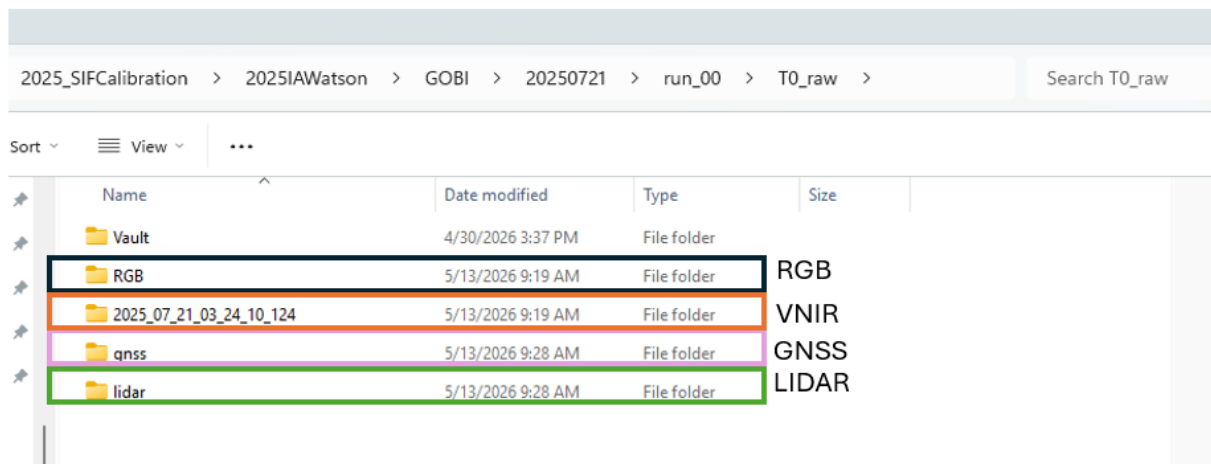
Important

Ensure the **Reflectance Target Location** parameter references the correct target values that correspond to your specific calibration panels.

2. Format the raw data

The example CALViS flight used here is **run_00**, a flight from USYD.

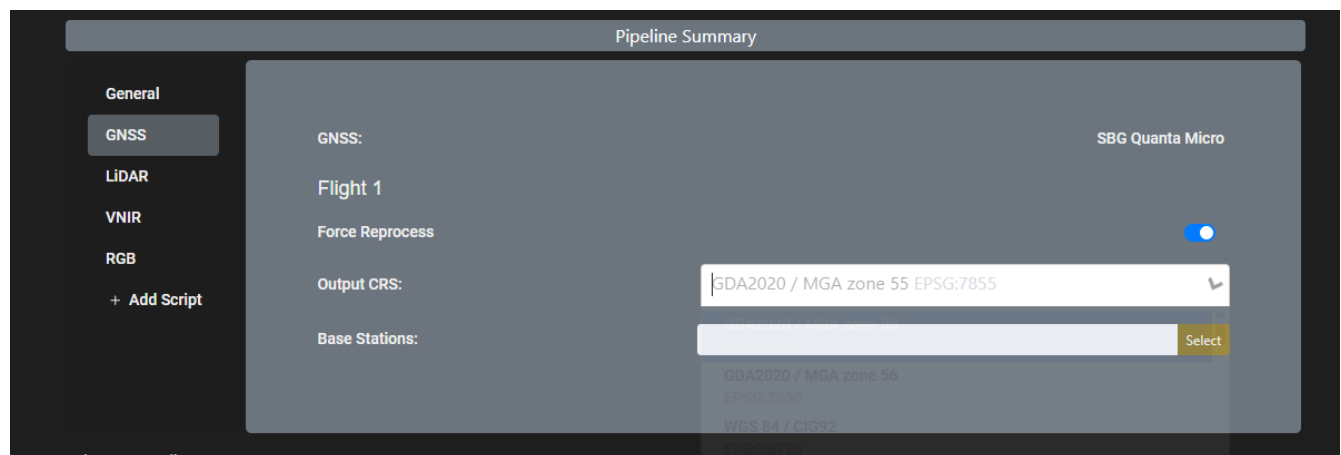
A complete CALViS dataset has four key components — **VNIR, RGB, LiDAR, and GNSS** — and these must be arranged correctly before a working graw can be bundled. The raw data folder will look like this:



3. Create the graw bundle

Pre-bundle checks Before each graw, confirm you have:

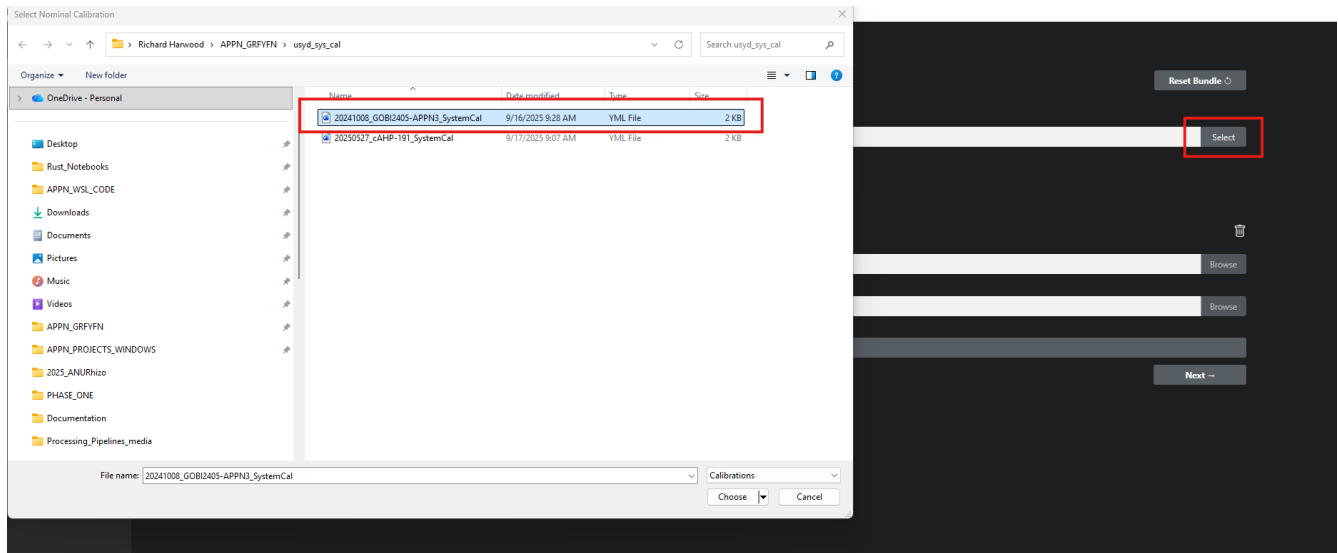
- ☐ System calibration file (matching the sensor used).
- ☐ Radiometric calibration files.
- ☐ Reflectance target file.



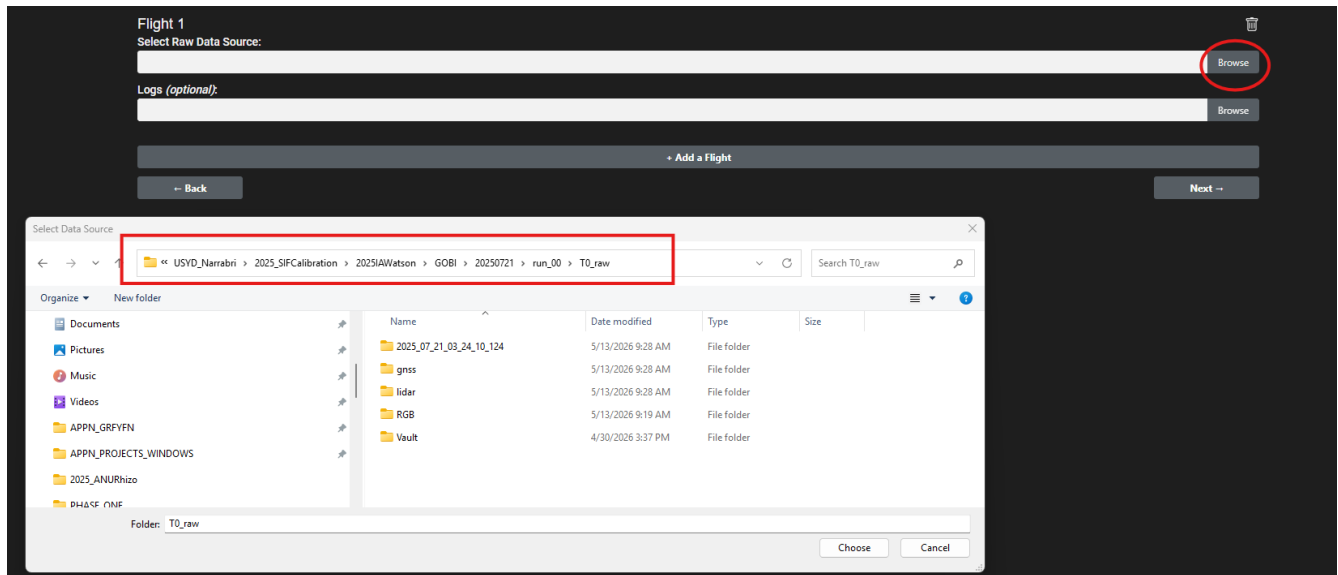
Note here we switched from UOA to USYD

Bundling steps

1. Set the system calibration file to match the sensor you are using.



2. Choose the raw data path.



3. Click *Next* to view the optional parameters.

Note

Unless you have a specific reason to set an extent, **leave it blank** — data can be cropped to the hyperspectral capture area later, which is ideal for most standard flights. Other fields here are also optional.

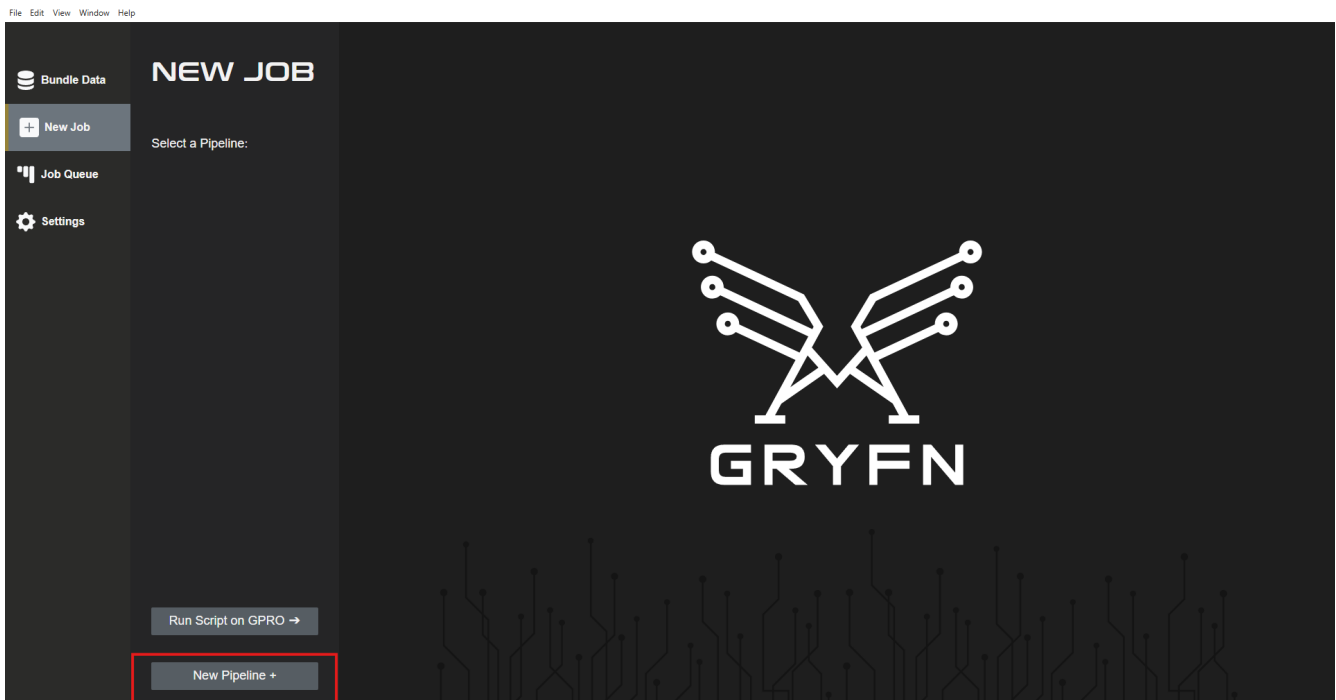
4. Click **Next**. Give the bundle a logical name and save it under `T0_raw`.

5. Click **Create bundle**.

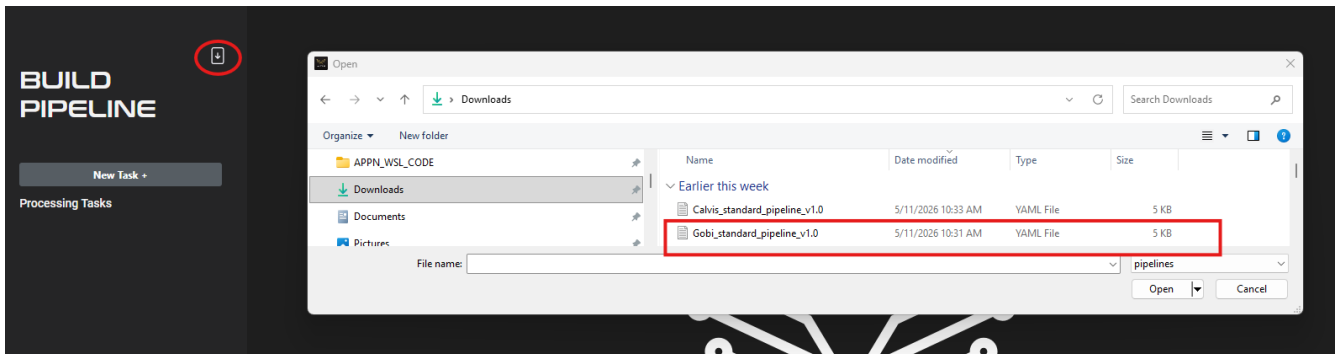
4. Load the GOBI_standard_pipeline

Import the standard GOBI pipeline YAML (`Calvis_standard_pipeline_v1.0.yaml`)

1. Open the pipelines manager.



2. Import the Gobi_standard_pipeline YAML.

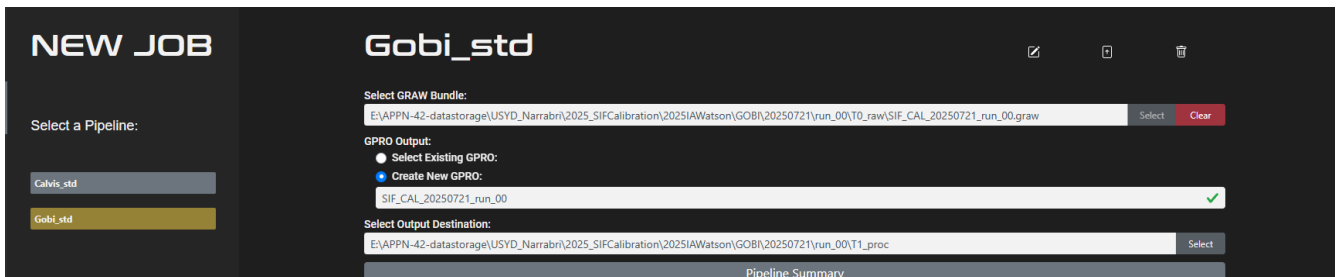


3. Confirm the imported pipeline is listed.



5. Run the pipeline on a graw

1. Select **New Job** and choose **Gobi_std** from the pipelines list. Choose your graw, name your gpro, and set the gpro output location.



Important

The gpro **must** be saved under **T1_proc**.

2. Configure GNSS processing.

Select Output Destination:
 E:/APPN-42-datastorage/USYD_Narrabri/2025_SIFCalibration/2025IAWatson/GOBI/20250721/run_00/T0_raw Select

Pipeline Summary

General

GNSS: SBG Quanta Micro

Flight 1

Force Reprocess ☑

Output CRS: GDA2020 / MGA zone 54 EPSG:7854 x v

Base Stations: Select

+ Add Script

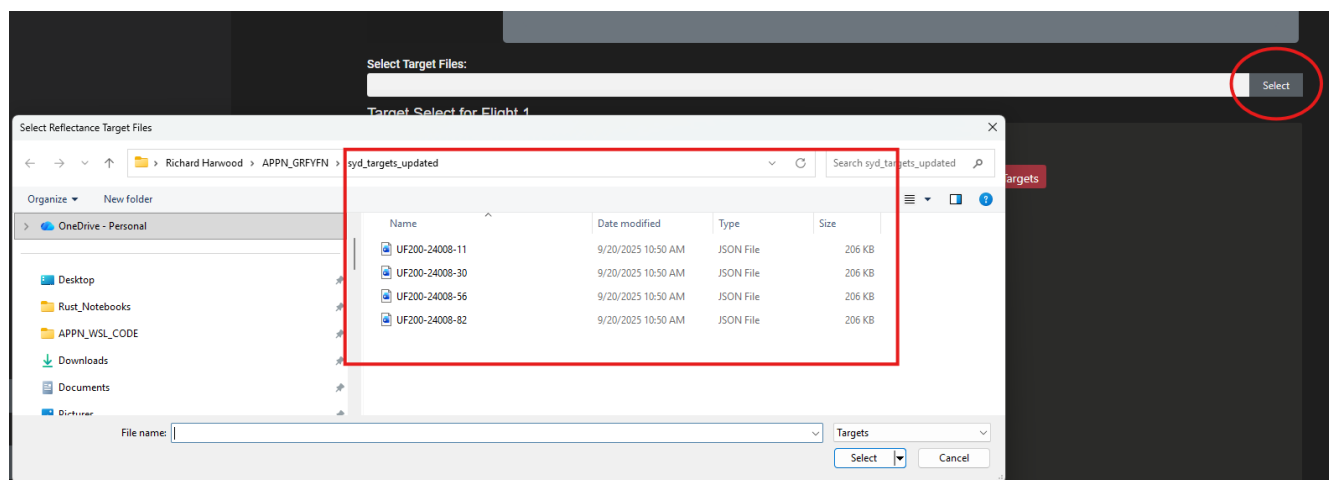
Select Target Files:

Caution

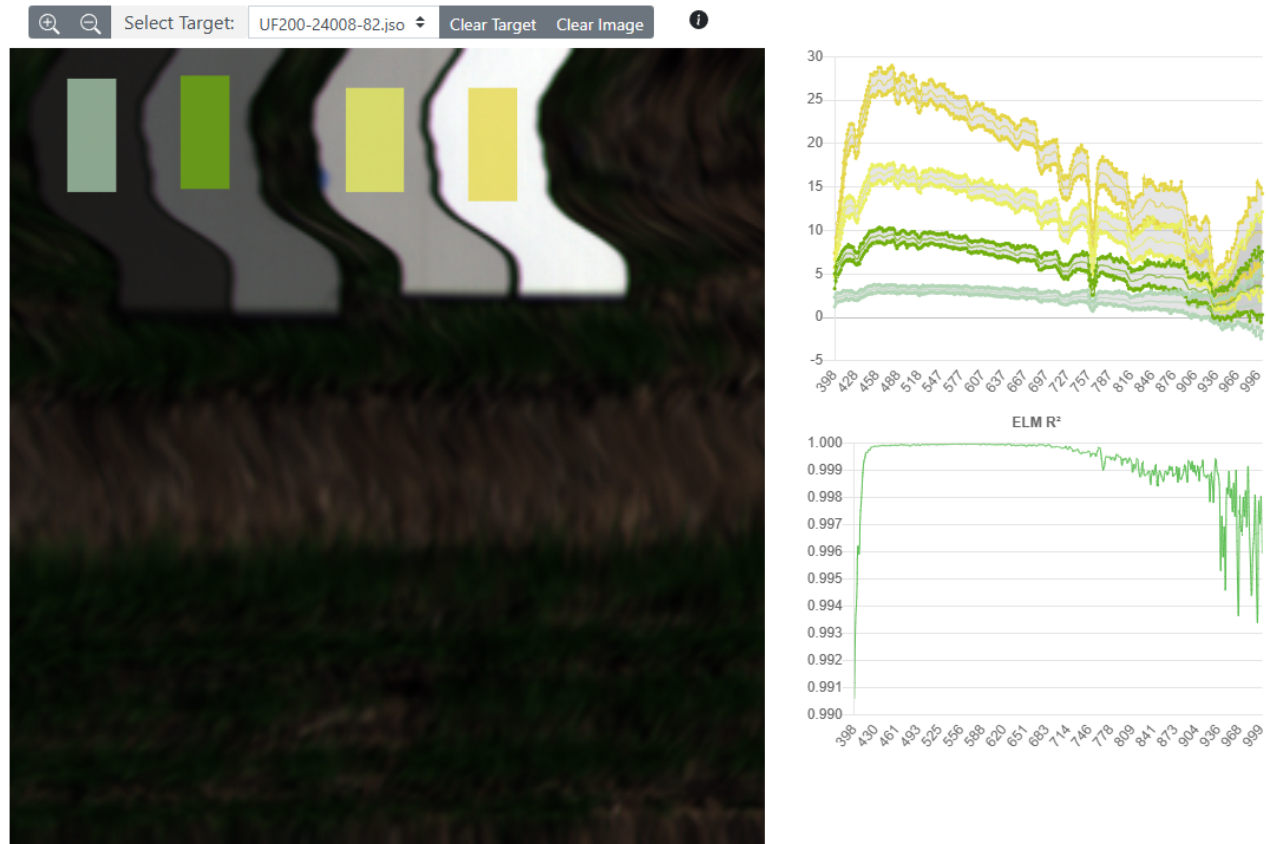
Double-check your **Datum**, **Grid**, and **Zone** before continuing. Errors here propagate through every downstream product.

There is no PPRTX option for GOBI.

3. Load the reflectance target files. For this example flight we use the UOA panel target files.



4. **VNIR ELM**. Cycle through the images until you find the four panels, then click **Draw target bounds**. (note explained in greater detail in the CALViS steps above)



[!IMPORTANT] make sure to note that the Y-axis in the top right box is showing radiance units and not raw counts (Should be obvious, ranges in the tens/hundreds vs thousands/tens of thousands)]

GOBI outputs

The GOBI standard pipeline produces the LiDAR, hyperspectral, and RGB products listed below. See Standard Data Products for the canonical specifications, file sizes, and software compatibility.

Product	Output filename	Resolution	Format	Notes
LiDAR Digital Surface Model (DSM)	LiDAR_DSM.tif	8 cm (fixed)	GeoTIFF	Extent: VNIR scene
LiDAR Digital Terrain Model (DTM)	LiDAR_DTM.tif	1 m	GeoTIFF	Extent: processing extent
Combined LiDAR Point Cloud	LiDAR_CombinedPointCloud.las	Native point spacing	LAS	Outliers removed during combination
VNIR Orthomosaic	VNIR_Orthomosaic.bin	(GSD-based)	ENVI (.bin + .hdr)	binning = 2, radiometric calibration applied
RGB Orthomosaic	RGB_Orthomosaic.tif	Use Gryfn flight calculator	GeoTIFF	Feature-matching (SIFT) bundle adjustment applied